



Assignment-1

Web Engineering

➤ Submitter By	:	Muhammad Essa
➤ Submitter To	:	Mr Azhar ALi
➤ Subject	:	Web Engineering
➤ Semester	:	6 th
➤ Section	:	(A)
➤ Department	:	BSCS
➤ Registration No	:	2023-KIU-BS5215

<https://github.com/MuhammadEssa-514/Web-Engineering-Course>

visit above link for repository to check assignment code

<https://muhammadesa514.vercel.app>

visit portfolio

I did this assignment in three ways .

Assignment 1 - Web Engineering

Teacher: Mr Azhar Ali

Submitted By: Muhammad Essa

[Github Repo Link](#)

[Portfolio](#)

❖ Html Basic Structure of boiler plate

❖ `<!DOCTYPE html>`

- ❖ This is a mandatory declaration at the very top of the document that tells the browser the page is written in HTML5. If we don't include this declaration, the browser will assume the page is written in an older version of HTML.

❖ `<html>`

- ❖ The html tag is the root element because it acts as a container for other elements and tells the browser everything inside this tag is html

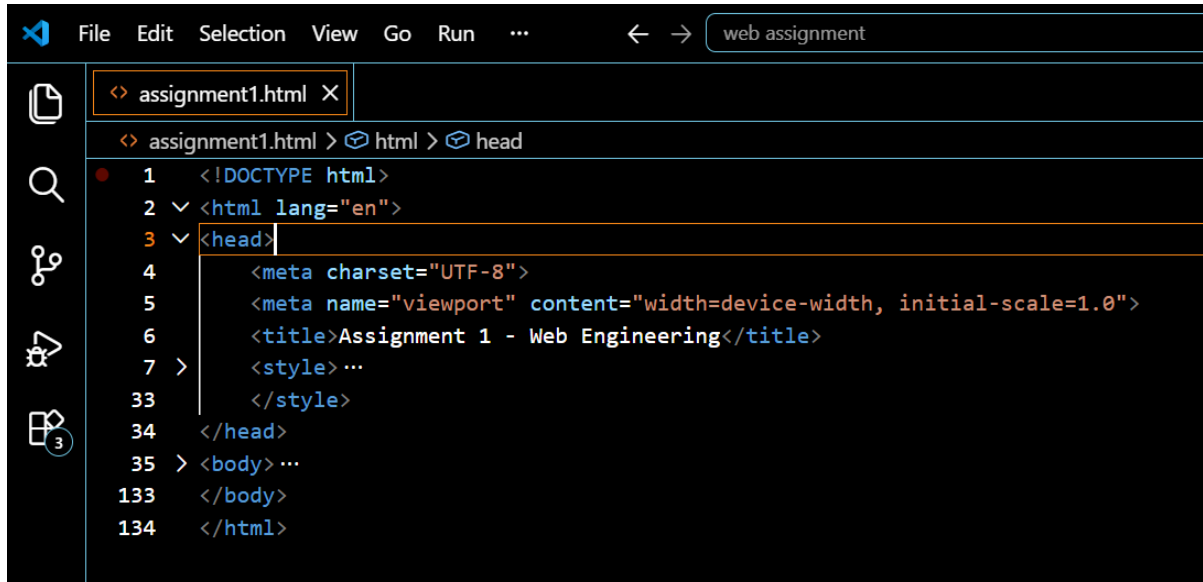
❖ `<head>`

- ❖ A container for metadata (data about the page). This information is not visible to users but is crucial for browsers and search engines.

❖ `<body>`

- ❖ This contains all the visible content, such as text, images, buttons etc.

❖ IDE SS Assignment with simple Css for clear tags visibility



```
File Edit Selection View Go Run ... web assignment
< assignment1.html X
assignment1.html > html > head
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Assignment 1 - Web Engineering</title>
7   <style> ...
33  </style>
34 </head>
35 <body> ...
133 </body>
134 </html>
```

❖ 2-Html Web Structure

- **Navbar** contain main links of pages for switching or redirecting to other page or web.
- **Body.** Contain all other content which is visible for users like text,image,videos and buttons etc.
- **footer.** botton section of web contain social links ,maps ,contact info, copy right 2notice and privacy policies.

❖ Check basic web structure visit repo for checking practically

HTML Web Structure

Top part of website

HomeAboutContactSignInSignUp

Html Basic Structure of boiler plate

<!DOCTYPE html>

This is a mandatory declaration at the very top of the document that tells the browser the page is written in HTML5.If we don't include this declaration, the browser will assume the page is written in an older version of HTML.

<html>

The html tag is the root element because it acts as a container for other elements and tells the browser everything inside this tag is html

<head>

A container for metadata (data about the page). This information is not visible to users but is crucial for browsers and search engines.

<body>

This contains all the visible content, such as text, images, buttons etc.

Footer Of Website

❖ Semantic Elements

- ❖ **Semantic Elements**
- ❖ Semantic elements are those elements that clearly describe their meaning and purpose for both the browser and the developer

❖ Check SS of Semantic element visit repo for checking practically

Semantic Elements

Semantic elements are those elements that clearly describe their meaning and purpose for both the browser and the developer

Header
Main Content
footer
section
paragraph
button

❖ Block Elements

❖ Block Elements:

A block-level element is a type of element that structures a web page by taking up the full width available and always starting on a new line.

Block Elements

A block-level element is a type of element that structures a web page by taking up the full width available and always starting on a new line

<section></section>
<header></header>
<footer></footer>
<main></main>
<nav></nav>
<form action=""></form>
<table></table>
<hr>

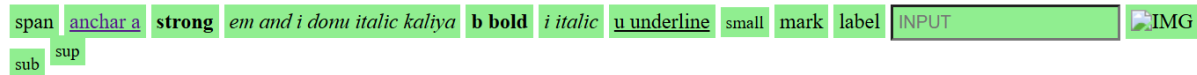
❖ Inline Elements

❖ Inline Elements:

- ❖ An inline element is a type of element that does not start on a new line and only takes necessary space for its content

Inline Elements

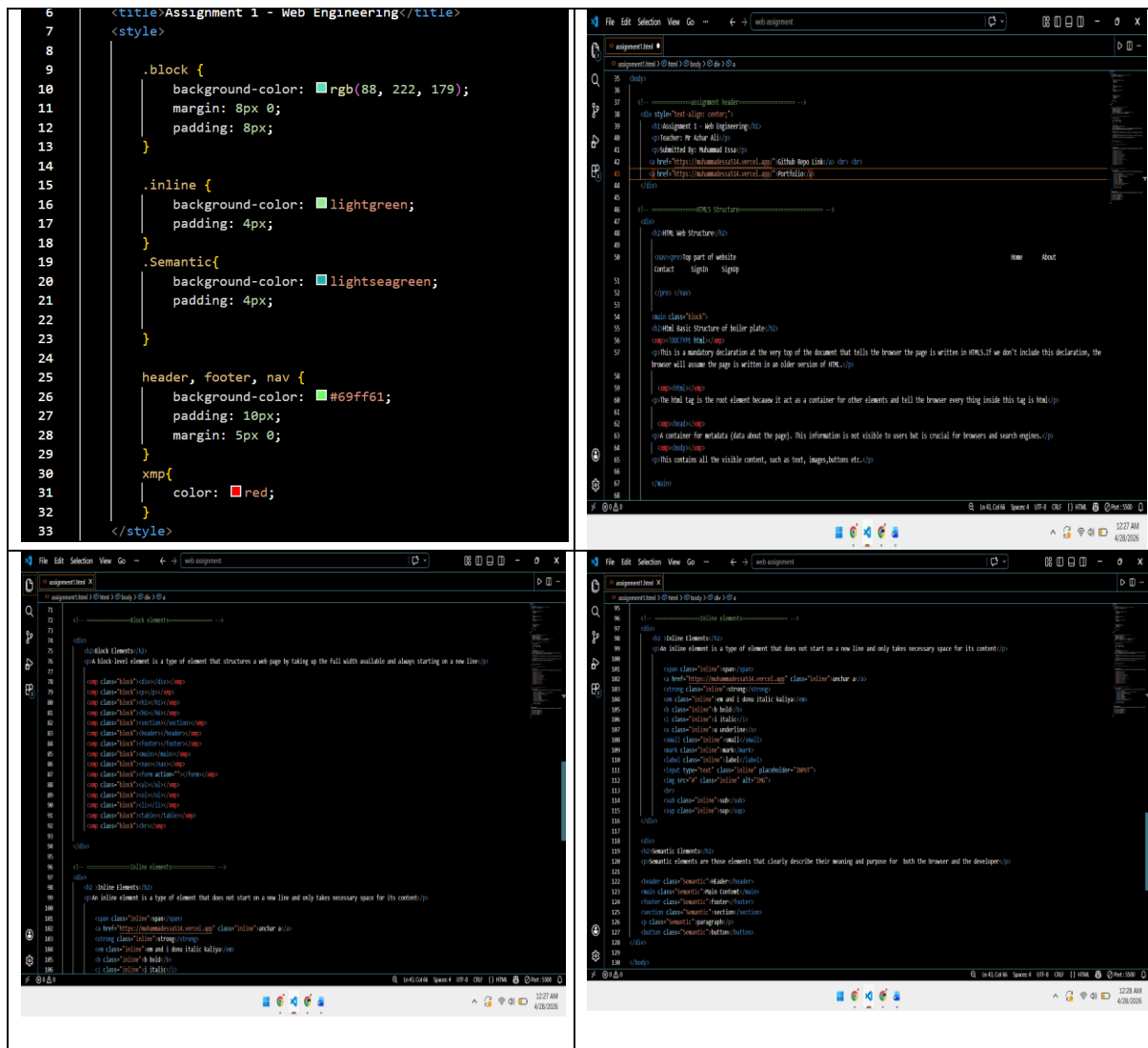
An inline element is a type of element that does not start on a new line and only takes necessary space for its content



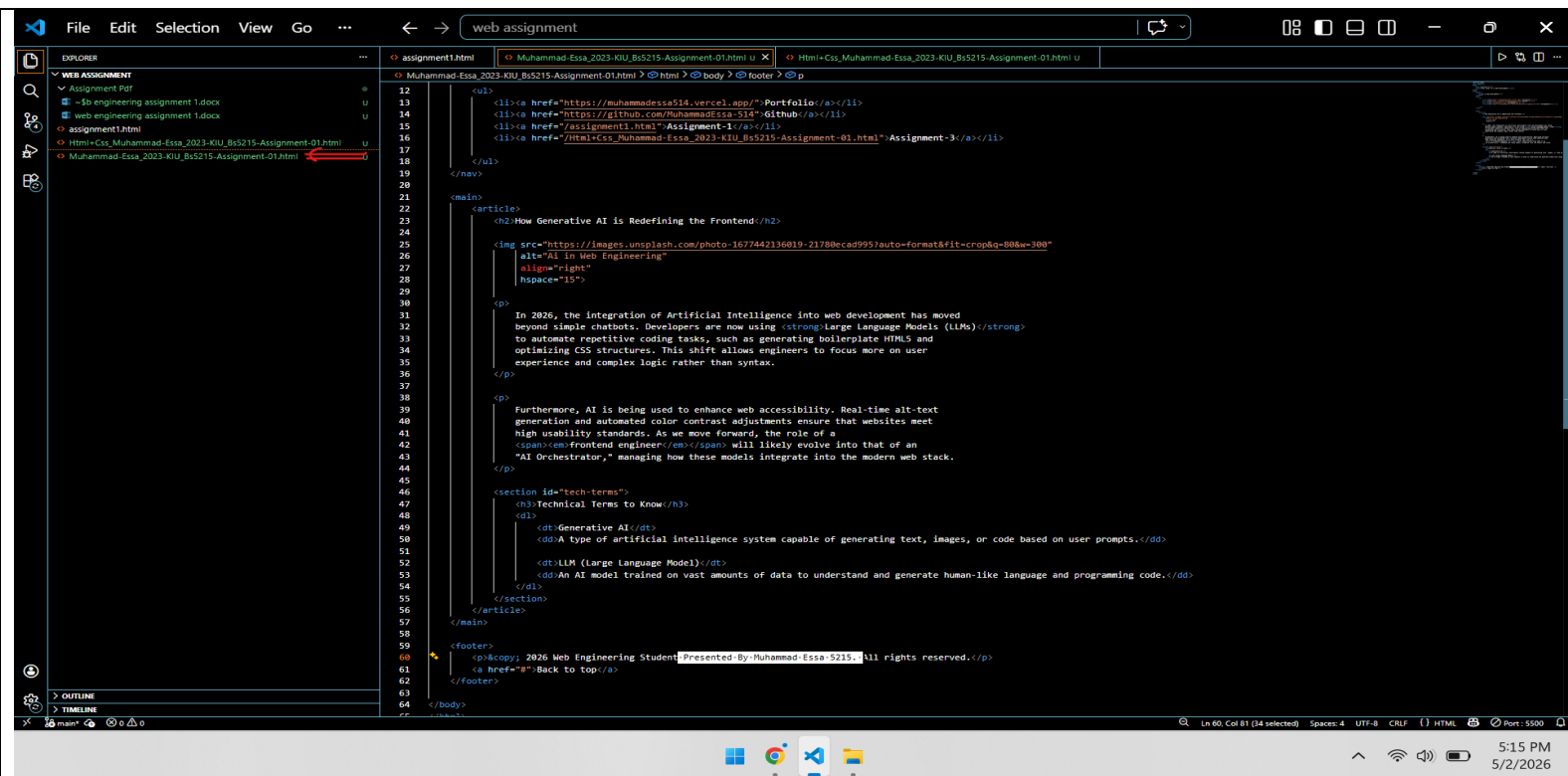
Semantic Elements

Semantic elements are those elements that clearly describe their meaning and purpose for both the browser and the developer

Assignment 1 with Simple Css-code

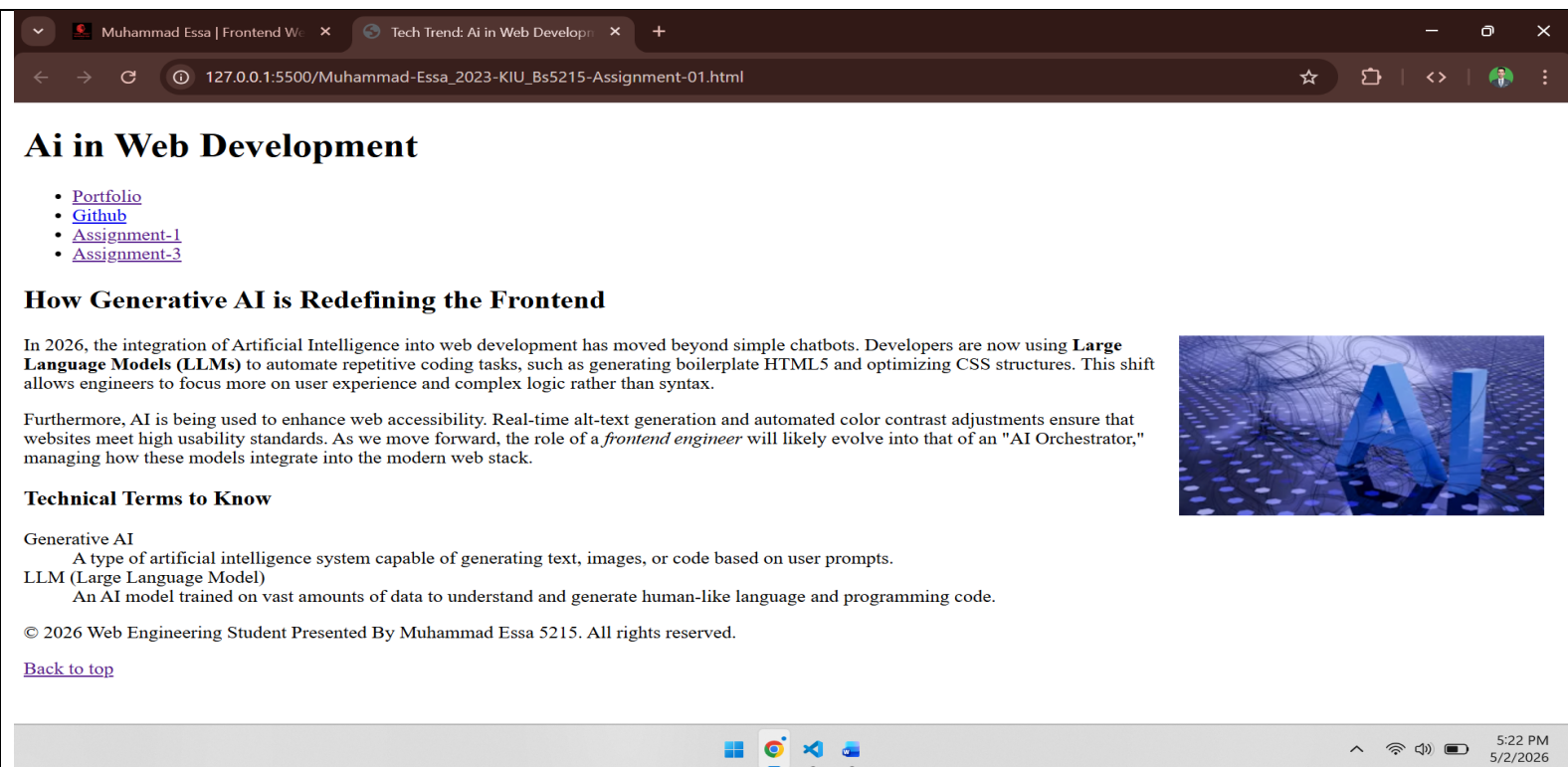


=====Assignment 1 without Css-code=====



```
12 <ul>
13 <li><a href="https://muhammedessa514.vercel.app/">Portfolio</a></li>
14 <li><a href="https://github.com/MuhammadEssa-514">Github</a></li>
15 <li><a href="assignment1.html">Assignment-1</a></li>
16 <li><a href="/html+css+Muhammad-Essa-2023-KIU_Bs5215-Assignment-01.html">Assignment-3</a></li>
17 </ul>
18 </nav>
19
20 <main>
21 <article>
22 <h2>How Generative AI is Redefining the Frontend</h2>
23
24 
28
29 <p>
30 In 2026, the integration of Artificial Intelligence into web development has moved
31 beyond simple chatbots. Developers are now using <strong>Large Language Models (LLMs)</strong>
32 to automate repetitive coding tasks, such as generating boilerplate HTML5 and
33 optimizing CSS structures. This shift allows engineers to focus more on user
34 experience and complex logic rather than syntax.
35 </p>
36
37 <p>
38 Furthermore, AI is being used to enhance web accessibility. Real-time alt-text
39 generation and automated color contrast adjustments ensure that websites meet
40 high usability standards. As we move forward, the role of a
41 <span><em>frontend engineer</em></span> will likely evolve into that of an
42 "AI Orchestrator," managing how these models integrate into the modern web stack.
43 </p>
44
45 <section id="tech-terms">
46 <h3>Technical Terms to Know</h3>
47
48 <dl>
49 <dt>Generative AI</dt>
50 <dd>A type of artificial intelligence system capable of generating text, images, or code based on user prompts.</dd>
51
52 <dt>LLM (Large Language Model)</dt>
53 <dd>An AI model trained on vast amounts of data to understand and generate human-like language and programming code.</dd>
54 </dl>
55 </section>
56 </article>
57 </main>
58
59 <footer>
60 <p><small>Copyright: 2026 Web Engineering Student Presented By: Muhammad Essa-5215. All rights reserved.</small></p>
61 <a href="#">Back to top</a>
62 </footer>
63
64 </body>
65 </html>
```

=Assignment 1 without Css-code Output=



AI in Web Development

- [Portfolio](#)
- [Github](#)
- [Assignment-1](#)
- [Assignment-3](#)

How Generative AI is Redefining the Frontend

In 2026, the integration of Artificial Intelligence into web development has moved beyond simple chatbots. Developers are now using **Large Language Models (LLMs)** to automate repetitive coding tasks, such as generating boilerplate HTML5 and optimizing CSS structures. This shift allows engineers to focus more on user experience and complex logic rather than syntax.

Furthermore, AI is being used to enhance web accessibility. Real-time alt-text generation and automated color contrast adjustments ensure that websites meet high usability standards. As we move forward, the role of a *frontend engineer* will likely evolve into that of an "AI Orchestrator," managing how these models integrate into the modern web stack.

Technical Terms to Know

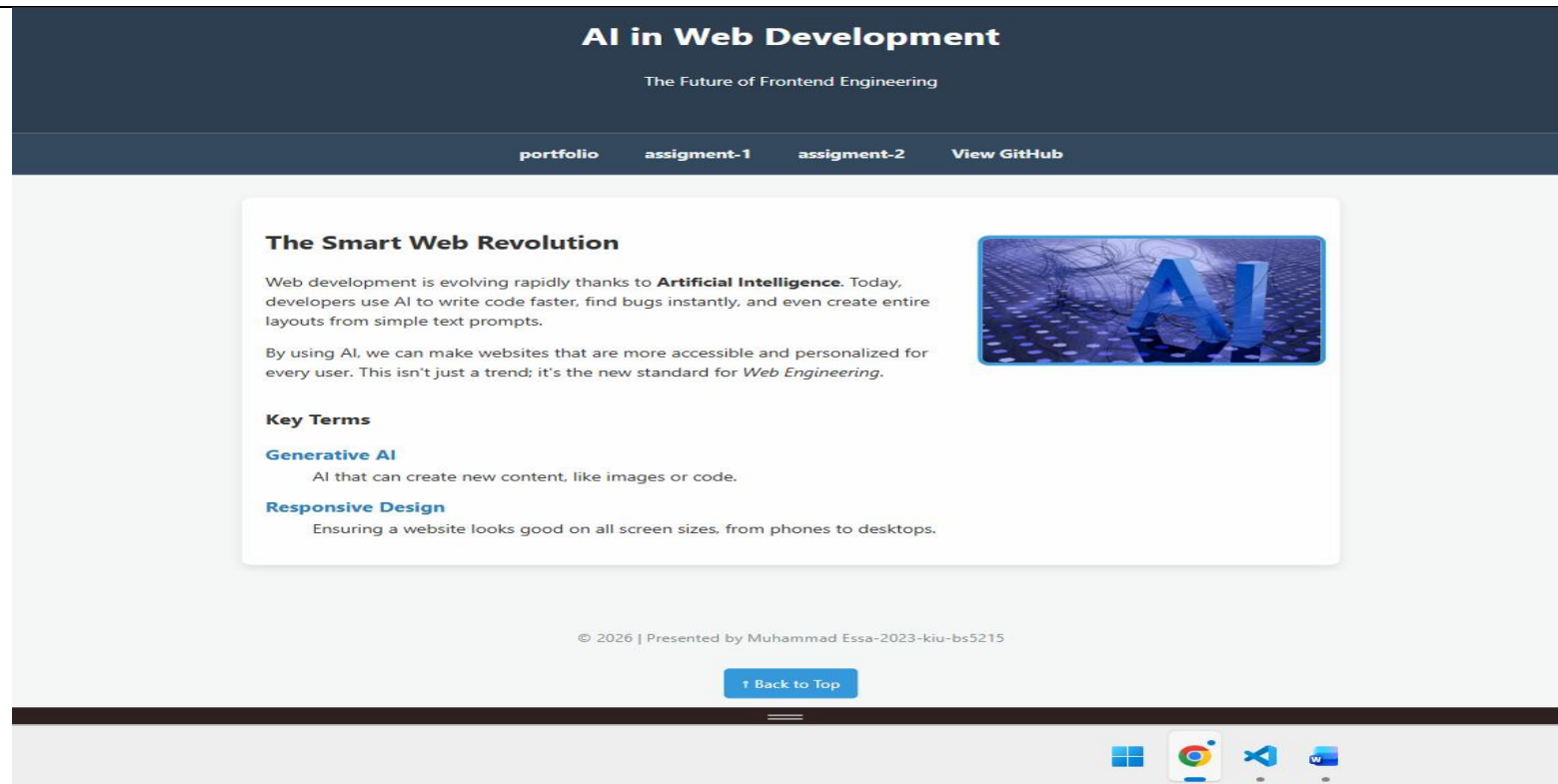
Generative AI
A type of artificial intelligence system capable of generating text, images, or code based on user prompts.

LLM (Large Language Model)
An AI model trained on vast amounts of data to understand and generate human-like language and programming code.

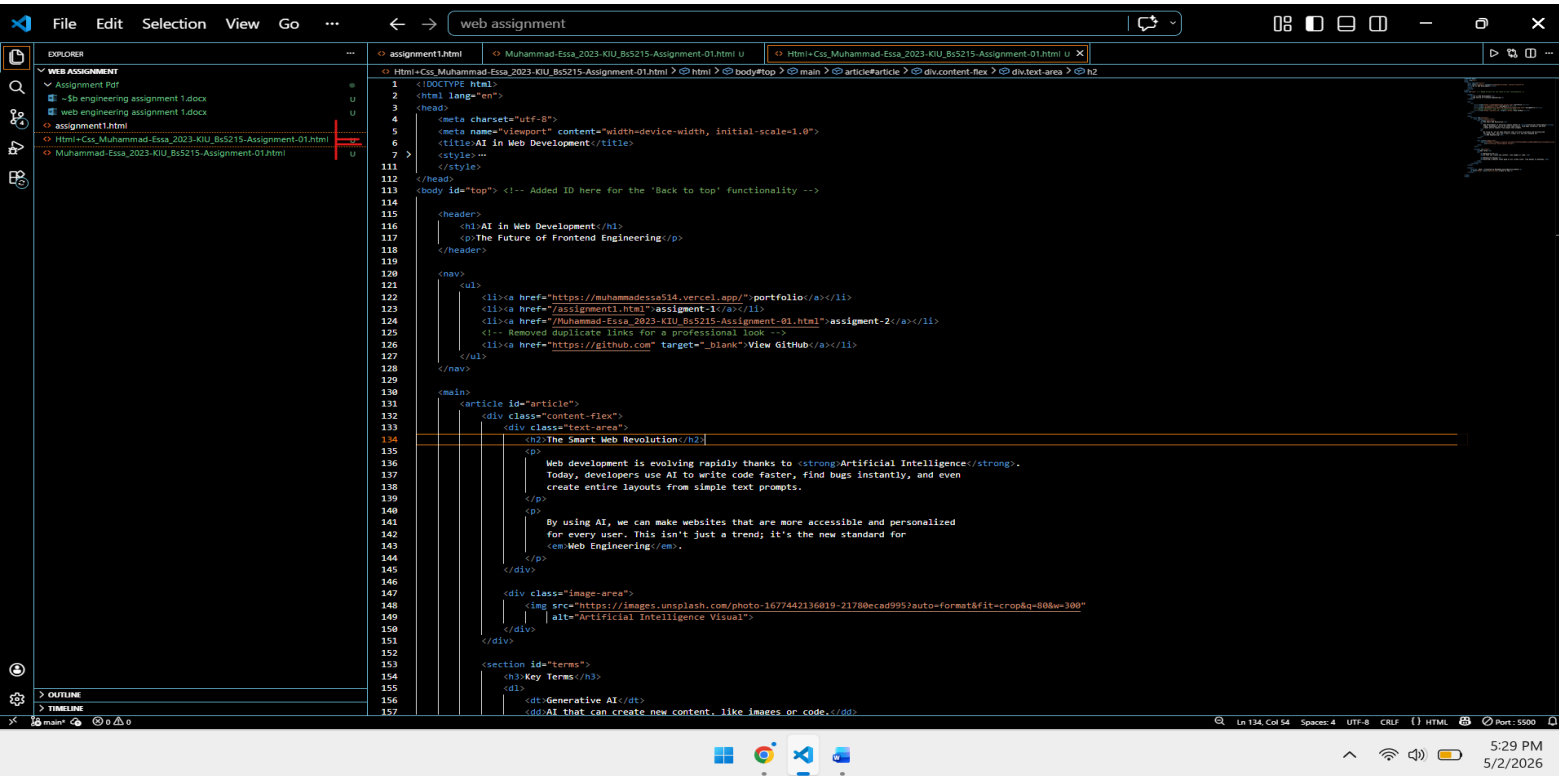
© 2026 Web Engineering Student Presented By: Muhammad Essa 5215. All rights reserved.

[Back to top](#)

===Assignment 1 with Proper Css-code===



=Assignment 1 with Proper Css Output=



Thanks Mr Azhar Ali